

KARTIK MISHRA

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SUMMARY

Machine learning engineer specializing in **real-time speech AI, low-latency inference, and production GenAI systems**. At Airtel, build and deploy end-to-end voice AI systems spanning **ASR fine-tuning, synthetic speech data generation, VAD/audio processing, GPU inference, and asynchronous serving**, powering a live broadband voice agent on Airtel IQ. Also built a production **RAG assistant over a 485K-LOC telecom codebase** and large-scale PySpark data platforms processing terabytes of daily 2G/4G/5G network data.

EDUCATION

Amity University, Noida, India
B.Tech. Biotechnology

Aug 2017 – Mar 2021
CGPA: 8.15/10

WORK EXPERIENCE

BHARTI AIRTEL

Machine Learning Engineer

Gurugram, India
Aug 2025 – Present

- Built and deployed a **production real-time voice agent** for broadband field diagnostics, orchestrating VAD → STT → LLM → TTS turn-taking in LangGraph; achieved **sub-1,000 ms round-trip latency** on Airtel IQ telephony while supporting concurrent live calls.
- Built a **60+ hour synthetic Hindi-English code-switched telecom speech corpus** using Indic TTS across diverse speaker personas, intents, and domain-specific utterances; developed a production-matched degradation pipeline with G.711 μ -law encoding, 8 kHz band-limiting, and probabilistic real-noise augmentation at 10–28 dB SNR to close the acoustic gap between training and telephony inference.
- Fine-tuned **NVIDIA Nemotron's cache-aware streaming RNN-T ASR** on synthetic + real telecom speech, achieving **7% English / 14% Hindi WER** on code-switched telephony; deployed the model on GPU via sherpa-onnx for low-latency streaming inference.
- Fine-tuned **Meta Omnilingual ASR 1B/3B LLM variants** on the same telecom-domain corpus, achieving **9% English / 16% Hindi WER** on code-switched telephony; deployed Omnilingual ASR on NVIDIA Triton for production inference and evaluated model/serving trade-offs across architectures.
- Owned the **production speech-inference stack end to end**, integrating Silero VAD, DeepFilterNet denoising, bandwidth extension, 8 kHz telephony preprocessing, FastAPI/WebSockets streaming, and a MongoDB-backed priority queue + call-state machine for concurrent live-session orchestration and low-latency audio processing.
- Built and deployed a **production RAG assistant for Airtel's Duct and VoLTE Steering systems**, indexing a **485K-LOC telecom codebase into 17K AST-aware function chunks** across 25 circles/states alongside 300+ Confluence documents; implemented BGE-M3 embeddings, Qdrant retrieval, BGE-reranker-v2-M3, and a self-hosted Qwen LLM on H100/vLLM to deliver streaming, grounded, source-cited answers to engineers.
- Rebuilt Airtel's **2G/4G/5G telecom data platform**, migrating 20+ PySpark/pandas ETL workflows to Dockerized Airflow with CI/CD and processing terabyte-scale daily broadband data; improved pipeline throughput by **75%** while reducing compute cost by **85%**, powering network-health KPIs, churn signals, and CX analytics.
- Led **nationwide Duct operations across 12+ states** as technical lead and primary SME/POC for a business-critical program impacting customer network performance, traffic, and tower/cell operations; coordinated cross-functional teams and deployed real-time monitoring that eliminated manual inspections and provided stakeholders with live operational visibility.

MOODY'S INSURANCE SOLUTIONS

Associate Analytics, Model Analyst

Noida, India
Apr 2023 – Aug 2025

- Architected an end-to-end data orchestration framework on Dagster, integrating high-frequency weather data from 30+ global sources with CI/CD automation via Jenkins and GitHub Actions, **improving pipeline efficiency by 90%**.
- Implemented a Django-based data visualization dashboard using Mapbox, JavaScript, MongoDB, and Python, enabling global wind station tracking and cross-referencing with the Postgres database to **identify over 18,000 missing stations**.
- Collaborated with the Event Response team to **automate Support Center uploads via an API**, reducing manual upload time from 40–60 minutes to **zero** and enhancing efficiency for natural disaster reports and forecasts.
- Unlocked visibility into HWind processes by building an **operational statistics dashboard** using JavaScript, Python, and Django, tracking daily and weekly data-feed counts and identifying problematic transactions and feed downtimes.

VERISK ANALYTICS (WOOD MACKENZIE)

Gurugram, India

Data Associate

Nov 2021 – Apr 2023

- Enhanced **data collection and processing pipelines in Python on AWS SageMaker**, improving efficiency by **60%**.
- Built Python automation to extract oil-price data from country reports, update Excel deliverables, and trigger email notifications on completion, **reducing end-to-end processing time by 90%**.
- Led **project handover sessions and roadmap execution** across teams, managing story allocation and task division using JIRA.
- Crafted a Power BI dashboard to **monitor JIRA workload and completion statistics**, improving team accountability and visibility into task fulfillment.

PUBLICATIONS

- Identification of vaccine targets & design of vaccine against SARS-CoV-2 coronavirus using computational and deep learning-based approaches - PeerJ, May 2022
- Identification of vaccine targets in pathogens and design of a vaccine using computational approaches - Nature Scientific Reports, September 2021
- To Study the Effect of Unconventional Treatment Protocol on COVID-19 Patients in Delhi Using Artificial Intelligence Based Methods - SSRN, September 2020

EXTRACURRICULAR ACTIVITIES

Member & AVP Education, Toastmasters Club, West Delhi - Organized weekly meetings, speaker selection, and onboarded new members (Apr 2023 – Mar 2025)

Volunteer & Drive Lead, Robin Hood Army, Delhi - Led 30+ food distribution drives and taught 25+ students weekly (Oct 2022 – Present)

Coordinator, CRC Placement Cell, Amity University - Orchestrated 100+ on-campus placement drives and pre-placement prep sessions (Feb 2019 – Mar 2021)

SKILLS

Languages: Python, SQL

Speech & ASR: NVIDIA NeMo, Nemotron, FastConformer RNN-T, Meta Omnilingual ASR (1B/3B), Whisper, Indic TTS, sherpa-onnx, Silero VAD, DeepFilterNet, PyTorch Audio

ML & GenAI: PyTorch, HuggingFace Transformers, PEFT/LoRA, LangGraph, LangChain, RAG, BGE-M3, BGE-reranker-v2-m3, AST-based code retrieval

Serving & Inference: NVIDIA Triton, vLLM, ONNX Runtime, FastAPI, WebSockets, Docker, CUDA, GPU inference, Streaming Inference

Data & Orchestration: PySpark, Pandas, Airflow, Delta Lake, ETL/ELT, CI/CD, Jenkins, GitHub Actions

Vector Search & Databases: Qdrant, MongoDB, PostgreSQL

Infrastructure & Tools: Linux, Git, AWS SageMaker, Streamlit